

How effective is music therapy in improving symptoms and quality of life in Parkinson's Disease

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Abstract

This independent literature review examined the effectiveness of music therapy in improving the symptoms and quality of life in people with Parkinson's disease, a progressive neurodegenerative condition that has significant and life-changing motor and non-motor symptoms. This investigation drew on more than 50 sources, including systematic reviews, meta-analyses, clinical trials and research to assess how music-based therapies impact areas such as gait, speech, emotional wellbeing and cognitive function.

Evidence shows that music therapy, especially Rhythmic Auditory Stimulation (RAS) can improve motor function, including walking velocity, stride length and gait. Non-motor benefits were observed in improved mood, reduced anxiety, and better social interaction.

The review critically evaluates studies and findings and outcomes in real world applications of music therapy. While benefits are well documented and established, more research is required to standardise therapy protocols and explore long-term outcomes.

The findings suggest that music therapy is safe, enjoyable and provides benefits. It can also serve as an effective complement and adjunct to traditional pharmacological treatments and other physical therapies.

Introduction

Parkinson's Disease is a progressive neurological condition that affects movement and is characterised by tremors, muscle stiffness, slow movement (bradykinesia), and involuntary movement (dyskinesia). It is thought to impact approximately 11 million people globally¹ and 166,000 in the UK².

It is a chronic and progressive condition that worsens over time, damaging the cells in the substantia nigra, a brain structure that is part of the Basal Ganglia and aids control of movement. The degeneration of these nerve cells inhibits the production of dopamine, a crucial neurotransmitter for movement control. The dopamine deficiency leads to motor and non-motor symptoms. The onset of Parkinson's Disease is believed to be a combination of environmental and genetic factors with incidences increasing with age. Environmental factors include head injuries, exposure to pesticides, industrial solvents, and air pollution³. Genetic factors other than age include biological sex, men being twice as likely to develop the disease, having close relatives with Parkinson's and specific gene mutations commonly in the LRRK2 and SNCA genes.

After speaking with a friend with Parkinson's Disease (PD), I was immediately interested by its idiopathic nature and wondered how patients with the disease use management techniques to aid their daily lives. As a person with a keen interest in neuroscience, I was fascinated by the disease and wanted to research psychological and physical techniques to aid brain function, particularly using music to facilitate neural plasticity. As a musician, I desired to know whether I could learn strategies to help patients manage and improve their symptoms using music and rhythm. As there is no cure for Parkinson's Disease, I explored music therapy as a non-pharmacological option that could potentially help manage symptoms and improve quality of life⁴.

What is music therapy?

Music therapy is a structured process using music-based interventions to help address emotional, cognitive, and physical systems. These interventions could include listening or playing instruments, singing, songwriting and improvisation. In this essay, I aim to evaluate the effectiveness of music therapy in Parkinson's Disease, looking deeper into the extent and ways in which it can improve symptoms, as well as the supporting

¹ Luo Y;Qiao L;Li M;Wen X;Zhang W;Li X; (no date) Global, regional, National Epidemiology and trends of parkinson's disease from 1990 to 2021: Findings from the global burden of disease study 2021, Frontiers in aging neuroscience. Available at: <https://pubmed.ncbi.nlm.nih.gov/39868382/#:~:text=Meth-ods:%20This%20study%20analyzed%20the,prevent%20and%20treat%20Parkinson's%20disease.> (Accessed: 17 August 2025).

²Parkinson's statistics (no date) Parkinson's UK. Available at: <https://www.parkinsons.org.uk/information/about-parkinsons/statistics> (Accessed: 17 August 2025).

³ Causes. Parkinson's Foundation. Available at: <https://www.parkinson.org/understanding-parkinsons/causes> (Accessed: 2 September 2025)

⁴ Complementary therapies (no date) Parkinson's UK. Available at: <https://www.parkinsons.org.uk/information/treatments/complementary-therapies> (Accessed: 10 September 2025)

evidence for improvement and the brain's ability to form new neural networks to improve brain function.

Neurological music therapy is typically used to help manage a vast range of conditions, including cancer, autism, anxiety, and mood disorders, learning disabilities and pain. Specifically, its application in Parkinson's Disease can directly improve balance and gait by using Rhythmic Auditory Stimulation (RAS) and can additionally help speech fluency through song. Other therapies, such as physiotherapy, help with strength and mobility training, occupational therapy with daily tasks, and speech and language therapy to address speech and swallowing issues that are commonly found with patients with PD.

The motor effect in PD affects non-motor symptoms. For example, nocturnal stiffness can lead to a lack of sleep, affecting emotional well-being, cognitive ability, and mood. Also motor fluctuations can trigger or worsen anxiety, depression. I aim to investigate how music therapy can help manage both motor and non-motor symptoms simultaneously.

Types of music therapy

Types of music therapy include receptive, improvisational, compositional, and recreational. Receptive music therapy is when the patient actively listens to music, selected by the therapist, which is either live or recorded, to help them relax and process emotions related to the stimuli. Improvisational therapy is when the patient and therapist interactionally work together to produce music following the therapist lead, acting as a form of nonverbal communication to express emotions. Similarly, compositional music therapy shows the interaction of sound between the patient and therapist, which leads to a piece of music created and built upon. Lastly, recreational music therapy is a range of music-based activities encouraging the patient to repeat phrases to the therapist. These categories cover a wide range of music therapies that often aid a variety of neurological conditions. Some of the main techniques used specifically to help Parkinson's Disease are: Rhythmic Auditory Stimulation (RAS) which is a type of Neurological Music Therapy (NMT), therapeutic singing, Therapeutic Instrumental Music Performance (TIMP), Patterned Sensory Enhancement (PSE) and music assisted relaxation. During this dissertation I aim to evaluate the different types of music therapy and how they can help the daily lives of patients by aiding specific symptoms.

Methodology

This independent narrative literature review examined the evidence on music therapy and music-based interventions for people with Parkinson's disease.

The literature search was conducted between August and December 2025 using Google and ResearchGate. The search began broadly with music therapy and its possible effects on physical function, psychological wellbeing and quality of life. It was then narrowed to research involving people with Parkinson's disease.

Searches combined terms relating to Parkinson's disease with terms such as music therapy, neurological music therapy, rhythmic auditory stimulation, singing, dance, gait, speech, cognition, mood and quality of life.

Systematic reviews and meta-analyses were prioritised because they brought together findings from several studies. Randomised controlled trials, controlled studies and selected observational research were also examined where they provided information about particular interventions, outcomes or possible neurological mechanisms.

Studies were considered relevant when they:

- involved people with Parkinson's disease;
- examined a music-based intervention;
- reported motor, non-motor or quality-of-life outcomes; or
- contributed relevant evidence about how music and rhythm may affect movement or brain function.

Research involving other conditions was only used when it provided relevant background about music, neurological function or rehabilitation. It was not treated as direct evidence of effectiveness in Parkinson's disease.

The evidence was evaluated by considering study design, sample size, the presence of a comparison or control group, outcome measures, intervention duration, follow-up, risk of bias and consistency with other studies. Particular attention was given to differences between short-term and long-term findings and between statistically significant results and clinically meaningful improvements.

Evidence both supporting and questioning the effectiveness of music therapy was included. Conflicting findings were compared, particularly for cognition, mood, quality of life and the duration of any benefits.

This was a narrative review rather than a formal systematic review. A predefined search protocol, database-specific search strategy and formal risk-of-bias assessment tool were not used. The selection of literature may therefore have been affected by the availability of full texts, search-engine ranking and the author's judgement about relevance.

The review should be understood as a critical synthesis of the literature located rather than an exhaustive account of every study in the field.

Literature Review

Music therapy has been increasingly explored as a complementary rehabilitation option for PD patients. There is a growing body of research investigating the effectiveness of this modality of therapy on motor and non-motor difficulties. Across the literature, the research demonstrates significant positive benefits of music therapy. However, the studies are heterogenous in methodology and showed some inconsistencies in which domains showed improvement and the extent to which some benefits were sustained.

The impact of Parkinson's Disease on quality of life

Symptoms of Parkinson's develop over time, Bradykinesia and Dyskinesia make daily life and tasks increasingly difficult, affecting balance sleep and memory. The physiological symptoms also take a toll on the psychological wellbeing of Parkinson's sufferers, often emerging as the disease progresses. As motor and non-motor symptoms simultaneously progress, patients lose their independence and ability to work and maintain an active lifestyle.

This review and discussion will consider the following domains to evaluate quality of life:

- Motor Symptoms: Physical functioning
- Non-Motor Symptoms: Including: Emotional/mental health, social participation, cognitive ability, independence, communication, sleep quality.

The decision was based on the domains used by PDQ-39⁵, a validated questionnaire of Quality of Life of PD patients.

⁵ R., P.V.C. (no date) PDQ-39: A review of the development, validation and application of a parkinson's disease quality of life questionnaire and its associated measures, Journal of neurology. Available at: <https://pubmed.ncbi.nlm.nih.gov/9617716/> (Accessed: 11 September 2025).

The use of Music Therapy

The Cleveland Clinic⁶ defines Music Therapy as “the use of music and/or elements of music (like sound, rhythm and harmony) to accomplish goals, like reducing stress or improving quality of life.” Specifically for patients with PD, music therapy is used to help a variety of the challenges, helping address motor symptoms alongside non-motor symptoms. As current pharmacological treatments do not address all aspects of PD, especially the psychological and non-motor symptoms, music therapy was explored by researchers and clinicians as a holistic approach to help manage symptoms that medication does not currently address and as a complementary therapy to conventional medical-based treatments.

Motor symptoms

Systematic reviews and meta-analyses can be one of the best ways to investigate quantitative research conducted obtained from clinical and Randomised Controlled Trials. Several RCTs highlighted the positive effect on motor symptoms of music therapy. A meta-analysis by Li et al⁷ which investigated ten randomised control trials looking at a scope of 529 participants found significant improvement to gait speed and stride length in PD patients who undertook Rhythmic Auditory Stimulation (RAS) based neurological music therapy. The limitations of this study are the heterogeneity, small samples, short follow-up, and language bias.

Lee and Ko further supported these findings⁸ by combining the results from 13 studies involving 417 participants, which showed a significant positive effect on the velocity of gait, stride length and improvement in mobility of PD patients. Interventions investigated in these studies included RAS and movement, active music therapy, vocal inter-

⁶ professional, C.C. medical (2025) What is music therapy, and how can it help me?, Cleveland Clinic. Available at: <https://my.clevelandclinic.org/health/treatments/8817-music-therapy> (Accessed: 11 September 2025).

⁷ Li, A. et al. (2025) Effectiveness and applications of neurologic music therapy in motor and non-motor rehabilitation for older adults with parkinson’s disease: A systematic review and meta-analysis, Frontiers. Available at: <https://www.frontiersin.org/journals/neurology/articles/10.3389/fneur.2025.1679881/full> (Accessed: 13 September 2025).

⁸ Lee, H. and Ko, B. (2023) Effects of music-based interventions on motor and non-motor symptoms in patients with parkinson’s disease: A systematic review and meta-analysis, International journal of environmental research and public health. Available at: <https://pmc.ncbi.nlm.nih.gov/articles/PMC9859027/> (Accessed: 10 September 2025).

ventions, and group-based music programs. These studies suggest that RAS interventions provided the most consistent positive results. These studies show strong agreement and confirm motor/non-motor benefits quantitatively.

One of the studies within the meta-analysis was a pilot study by Bukowska⁹ et al who investigated three types of neurological music therapy: TIMP, RAS, PSE and found significant increases in velocity, step length, stride length. The sample consisted of a control group of twenty-five and experimental group of thirty participants of members with PD who could walk independently. The controlled design strengthens causal inference, subject to sample size, allocation, blinding and other study limitations. The results also showed there was no deterioration in the reported gait or posture measures was observed. The study was too small to establish overall safety, however this could suggest a reduced risk of falls and harm.

A methods paper by Buard et al¹⁰ analysed an RCT using one hundred Parkinson's Disease patients randomly assigned to receive fifteen sessions of TIMP with rhythm or TIMP without rhythm or part of the control group over 5 consecutive weeks. Brain oscillators were used to measure the underlying mechanisms and to observe fine motor outcomes.

Due to basal ganglia circuit malfunction in patients with PD, their internal cue of rhythm is compromised. The Handbook of Neurologic Music Therapy¹¹ emphasises the importance of timing and rhythm to organise the motor systems within the brain. This book also considers how different types of music therapy may be more helpful for some compared to others and discussed many different interventions that have been tested. The findings and opinions expressed in this book align with the empirical evidence found in the meta-analyses and research studies. However, the book highlights that not all patients respond to RAS in the same way the approach must be tailored to each person. Furthermore, as short-term effects are promising music therapy should be consistent over an extended period of time.

⁹ Bukowska, A.A. et al. (2025) Neurologic music therapy training for mobility and stability rehabilitation with parkinson's disease – a pilot study, *Frontiers*. Available at: <https://www.frontiersin.org/journals/human-neuroscience/articles/10.3389/fnhum.2015.00710/full> (Accessed: 11 September 2025).

¹⁰ Buard I;Lattanzio L;Stewart R;Thompson S;Sjoberg K;Hookstadt K;Morrow M;Holden SK;Sillau S;Thaut M;Kluger B; (no date) Randomized controlled trial of neurologic music therapy in parkinson's disease: Research rehabilitation protocols for mechanistic and clinical investigations, *Trials*. Available at: <https://pubmed.ncbi.nlm.nih.gov/34454592/> (Accessed: 13 September 2025).

¹¹ Thaut, M.H. and Hömberg, V. (2025) *Handbook of Neurologic Music therapy*. S.I.: OUP Oxford.

Additionally, Pohl et al¹² examined the longevity of the alleviated symptoms during a 3 month follow up of his experiment and discovered objective measures did not show clear immediate or lasting improvements in dual-task ability, cognition, balance or freezing of gait. This study also examined an interdisciplinary approach. However, this test needed more investigation as it was unblinded and had a small sample size.

Hackney and Earhart¹³ showed that combining rhythm music with dance (Tango in particular) improved motor symptoms. Zhou¹⁴ 2021 also showed that music and movement therapy has benefit. This shows that Music Therapy can be paired with dance therapy to produce an effective outcome.

Non motor symptoms

Beyond motor effects, several studies investigated the usefulness of music therapy on non-motor symptoms. Music based systematic review by Sotomayor¹⁵ looked at 58 studies to demonstrate positive effects in mood, reduced anxiety and enhanced self-expression in multiple studies after experiencing active or passive music therapy, or both. The study specifically discusses how the positive effect on breathing, swallowing and speech directly helps improve communication which helps boost morale. This review has strong alignment with other reviews on motor/emotional benefits. However, there is no formal appraisal of bias, and the methodologies of therapy are heterogeneous.

A systematic review by Węgrzynek-Gallina¹⁶ compared 13 studies and found, the authors reported short term positive effects on mood and cognitive function. This review showed strong alignment with the previous review and that of Li and Lee and Ko

¹² Pohl, P. et al. (2020) Group-based music intervention in Parkinson's disease - findings from a mixed-methods study, Clinical rehabilitation. Available at: <https://pmc.ncbi.nlm.nih.gov/articles/PMC7132435/> (Accessed: 20 September 2025).

¹³ GM;, H.M. (no date) Effects of dance on movement control in Parkinson's disease: A comparison of Argentine Tango and American ballroom, Journal of rehabilitation medicine. Available at: <https://pubmed.ncbi.nlm.nih.gov/19479161/> (Accessed: 21 September 2025).

¹⁴ Zhou Z;Zhou R;Wei W;Luan R;Li K; (no date) Effects of music-based movement therapy on motor function, balance, gait, mental health, and quality of life for patients with parkinson's disease: A systematic review and meta-analysis, Clinical rehabilitation. Available at: <https://pubmed.ncbi.nlm.nih.gov/33517767/> (Accessed: 21 September 2025).

¹⁵ Machado Sotomayor, M.J. et al. (2021) Music therapy and parkinson's disease: A systematic review from 2015-2020, International journal of environmental research and public health. Available at: <https://pmc.ncbi.nlm.nih.gov/articles/PMC8582661/> (Accessed: 22 September 2025).

¹⁶ Węgrzynek-Gallina et al. (2024) Music and Parkinson's disease: the state of the art of music therapy in Parkinson's disease – a systematic review: <https://www.neurologia.com.pl/assets/pdf/artykuly/144-153-an-4-2024-wegrzynek-pdf.pdf>

However, there was also a lack of blinding for the participants and the outcome assessors, which introduces an elevated risk of bias within the findings. These randomised controlled trials had small sample sizes and heterogeneity of interventions, using different methods and types of music therapy makes the trials unstandardised.

Another 2023 systematic review of randomised controlled trials (RCTs) by Citon and Hamdan¹⁷, which only included three high-quality RCTs, concluded there was no clear evidence of efficacy for music-based interventions on cognitive outcomes.

Butala et al¹⁸ found that group-based weekly music/singing interventions improved measures of emotional well-being, body discomfort, and improvement in the communication domain. The therapy also improved social connection, allowing the patients to build confidence within their interactions and relieve feelings of isolation.

The meta-analysis by Li¹⁹ also explored the results for improvement on mood, cognition, and quality of life. The results were inconsistent and were not statistically significant. This outcome may be due to the short length of the RCT (ranging from 4-8 weeks), a longer study, may elicit greater long-term changes in mood and quality of life.

A small-scale single-blinded study by Spina²⁰ in 2016 showed that cognition (measured by cognitive flexibility, processing speed, attention, working memory) improved in patients following music therapy. However, the effect was not sustained when reviewed 6 months after the end of the therapy. Another small-scale study showed that singing, especially group singing, may provide benefit to maintain vocal skills and slow the vocal deterioration that follows Parkinson's Disease²¹.

¹⁷ Citon, L.F. and Hamdan, A.C. (2023) Effectiveness of music-based interventions for cognitive rehabilitation in parkinson's disease: A systematic review of randomized controlled clinical trials, *Psicologia, reflexao e critica : revista semestral do Departamento de Psicologia da UFRGS*. Available at: <https://pmc.ncbi.nlm.nih.gov/articles/PMC10415237/> (Accessed: 11 October 2025).

¹⁸ Butala A;Li K;Swaminathan A;Dunlop S;Salnikova Y;Ficek B;Portnoff B;Harper M;Vernon B;Turk B;Mari Z;Pantelyat A; (no date) Parkinsonics: A randomized, blinded, cross-over trial of group singing for motor and nonmotor symptoms in idiopathic parkinson disease, *Parkinson's disease*. Available at: <https://pubmed.ncbi.nlm.nih.gov/36247106/> (Accessed: 1 October 2025).

¹⁹ Li, A. et al. (2025) Effectiveness and applications of neurologic music therapy in motor and non-motor rehabilitation for older adults with parkinson's disease: A systematic review and meta-analysis, *Frontiers*. Available at: <https://www.frontiersin.org/journals/neurology/articles/10.3389/fneur.2025.1679881/full> (Accessed: 14 October 2025).

²⁰ Spina et al. Music therapy for motor and nonmotor symptoms of parkinson's disease: A prospective, randomized, controlled, single-blinded study - spina - 2016 - *journal of the American Geriatrics Society* - Wiley Online Library. Available at: <https://agsjournals.onlinelibrary.wiley.com/doi/10.1111/jgs.14295> (Accessed: 16 October 2025).

M, Tanner. Does singing and vocal strengthening improve vocal ability in people with parkinson's disease?: *Arts & Health: Vol 8 , no 3* - get access. Available at: <https://www.tandfonline.com/doi/full/10.1080/17533015.2015.1088047?scroll=top&needAccess=true> (Accessed: 11 December 2025).

Li-Hua et al²², in a mini review of music therapy noted that music may partially bypass defective dopamine production in the basal ganglia and stimulate the cerebellum, a part of the brain involved in movement.

Evidence for neural plasticity

Parkinson's Disease causes a reduction in dopamine which can affect movement and wellbeing²³. Due to the dopamine deficiency in the Basal ganglia, alternative neural pathways are required by the brain. A research paper by Raglio²⁴ highlights how music can activate the limbic, paralimbic, and prefrontal circuits in the brain. Repeated activation using techniques such as RAS creates strengthened synaptic pathways which are associated with mood regulation and executive function. This suggests that music may reorganise emotion and motor pathways, which are disrupted in PD. A 2018 review by Koshimori et al²⁵. proposed neurological changes in the brain because of music therapy.

EEGs (electroencephalogram)²⁶ conducted on PD patients show the entrainment of beta and gamma rhythms to external beats which helps compensate for the lack of internal rhythm. This evidence is objective and quantitative and supports the research on neural plasticity showing short term positive effects however evidence of long-term oscillatory reorganisation is suggestive but limited.

Discussion

Evaluation of the effect of music therapy on motor symptoms

The Bradykinesia and Dyskinesia²⁷, due to the degeneration of dopamine generating cells, in Parkinson's Disease means patients often have trouble with balance, posture, and maintaining a stable gait. The motor effects can cause other physical problems such as speech and swallowing which can make daily tasks increasingly difficult.

²² Li-Hua. et al. Precision medicine and music therapy for Parkinson's Disease
<https://www.sciencedirect.com/science/article/pii/S2590112525000866#b0055>

²³ Remy P;Doder M;Lees A;Turjanski N;Brooks D; (no date) Depression in parkinson's disease: Loss of dopamine and noradrenaline innervation in the limbic system, Brain : a journal of neurology. Available at: <https://pubmed.ncbi.nlm.nih.gov/15716302/> (Accessed: 24 October 2025).

²⁴ Raglio, A. (2015) 'Effects of music and music therapy on mood in neurological patients', World Journal of Psychiatry, 5(1), p. 68. doi:10.5498/wjp.v5.i1.68.

²⁵ MH;, K.Y. (no date) Future perspectives on neural mechanisms underlying rhythm and music based neurorehabilitation in parkinson's disease, Ageing research reviews. Available at: <https://pubmed.ncbi.nlm.nih.gov/30005957/> (Accessed: 24 October 2025).

²⁶ Waninger, S. et al. (2020) *Neurophysiological biomarkers of parkinson's disease*, Journal of Parkinson's disease. Available at: <https://pmc.ncbi.nlm.nih.gov/articles/PMC7242849/#:~:text=Results:,measured%20by%20dopamine%20transporter%20PET>. (Accessed: 24 October 2025).

²⁷ NHS choices. Available at: <https://www.nhs.uk/conditions/parkinsons-disease/symptoms/> (Accessed: 11 October 2025).

One of the most disabling symptoms is gait freezing and reduced stride length. Patients often have trouble initiating movement and maintaining a steady rhythm. Postural instability increases fall risk and loss of independence. There was increased support for the positive impact on motor symptoms because of Music therapy. Rhythmic auditory stimulation was most cited across the literature and provided the most consistent empirical evidence. Rhythmic Auditory Stimulation (RAS)²⁸ is a form of entrainment technique that uses regular beats using musical timing or metronomes. Specifically, when using RAS for gait training it is applied in the form of regular isochronous auditory pulses such as metronome clicks, or metrical acoustical beats embedded in instrumental music. Typically, when first used, it matches the patients' preferred cadence and then the speed is gradually increased or decreased by 5-10%²⁹. A strength of RAS is its positive short-term impacts, resulting in fast improvements even from the first session, which provide evidence of short-term improvements in gait velocity and stride length. A meta-analysis found that rhythmic auditory stimulation was associated with an average improvement in walking velocity of 0.12 metres per second (95% CI 0.07–0.16; $p < 0.00001$). The confidence interval suggests that the true average improvement was likely to fall between 0.07 and 0.16 metres per second. The low p-value indicates that the observed difference was unlikely to be explained by chance alone. However, statistical significance does not establish how meaningful the improvement would be in everyday life or remove concerns about study quality, heterogeneity and bias.

RAS is also easy to implement and cost effective, as it can be performed at home or in clinics with simple tools like metronomes or music playback³⁰. As well as helping stride and cadence RAS also helps with freezing of gait, a common symptom for patients. In a systematic review and meta-analysis³¹ regular RAS training over weeks led to a reduction in falls in PD patients. A limitation of the studies investigating the effectiveness of

²⁸ Rhythmic auditory stimulation (RAS)® | Handbook of Neurologic Music therapy | oxford academic. Available at: <https://academic.oup.com/book/59860/chapter-abstract/511395107?redirectedFrom=fulltext> (Accessed: 6 October 2025).

²⁹ Rhythmic auditory stimulation as a potential neuromodulator for parkinson's disease - sciencedirect. Available at: <https://www.sciencedirect.com/science/article/abs/pii/S1353802023001827> (Accessed: 6 October 2025).

³⁰ Forte, R., Tocci, N. and De Vito, G. (2021) The impact of exercise intervention with rhythmic auditory stimulation to improve gait and mobility in parkinson disease: An umbrella review, MDPI. Available at: <https://www.mdpi.com/2076-3425/11/6/685> (Accessed: 7 October 2025).

³¹ Ye X;Li L;He R;Jia Y;Poon W; (2022) Rhythmic auditory stimulation promotes gait recovery in parkinson's patients: A systematic review and meta-analysis, Frontiers in neurology. Available at: <https://pub-med.ncbi.nlm.nih.gov/35968291/> (Accessed: 7 October).

RAS is, that only short-term outcomes are observed as opposed to assessing daily functioning over extended periods³². Also, the impact on more severely affected patients who have advanced PD or severe gait/balance impairment is less well established. Furthermore, not all people with PD benefit equally, as some respond strongly (improved gait), others may show minor change.

Overall, research shows that RAS has a positive effect on improving mobility and gait for patients in Parkinson's to help aid them in their independence and daily tasks. Although research on long-term effects is limited, the short-term effects provided proves that music therapy should be very regular to maintain its effects.

Non-Motor Outcomes

Non-motor measures can be more difficult to quantify and may have varying degrees of impact on a sufferer's quality of life. The benefits are often subjective. Several studies have aimed to evaluate these areas in a variety of domains including cognitive function, mood and wellbeing, social connection, and sleep.

In addition to affecting movement, a loss of dopamine affects the mesolimbic and mesocortical pathways of the brain which are essential for emotional regulation³³. This can lead to, a decrease in motivation, lowered emotion and mood, reduced cognitive and memory function. In addition to reduced dopamine, Parkinson's Disease can impact on the serotonergic neurons in the brain resulting in reduced serotonin levels. This can result in depression, anxiety, sleep disturbances and fatigue³⁴.

Depression and anxiety can affect up to 50% of those with Parkinson's Disease³⁵. Depression is often undiagnosed and not always responsive to dopaminergic medication. Parkinson's Disease can impair attention, planning, working memory, this can affect daily functioning even in early onset of PD. Many patients develop hypophonia (quiet speech), monotone, or dysarthria which reduces their confidence and leads to social

³² Agarwal, A.S., Marwaha, A. and Sajid, A. (2005) The use of rhythmic auditory stimulation on gait parameters in parkinson's disease: A systematic review, Undergraduate Research in Natural and Clinical Science and Technology Journal. Available at: <https://www.urncst.com/index.php/urncst/article/view/410> (Accessed: 7 October).

³³ Remy P;Doder M;Lees A;Turjanski N;Brooks D; (2022) Depression in parkinson's disease: Loss of dopamine and noradrenaline innervation in the limbic system, Brain : a journal of neurology. Available at: <https://pubmed.ncbi.nlm.nih.gov/15716302/> (Accessed: 11 December 2025).

³⁴ C;, P.M. (2011) Serotonergic dysfunction in parkinson's disease and its relevance to disability, TheScientificWorldJournal. Available at: <https://pubmed.ncbi.nlm.nih.gov/22125431/>(Accessed: 11 December 2025).

³⁵ Jasinska-Myga, B. et al. (2010) Depression in parkinson's disease, The Canadian journal of neurological sciences. Le journal canadien des sciences neurologiques. Available at: <https://pmc.ncbi.nlm.nih.gov/articles/PMC3907778/#:~:text=The%20prevalence%20of%20psychopathology%20is,significantly%20worse%20quality%20of%20life.&text=Moreover%2C%20depression%20in%20PD%20of-ten%20remains%20underdiagnosed%20and%20undertreated.&text=All%20of%20these%20evidences%20strongly,longitudinally%20by%20a%20single%20clinician.> (Accessed: 2 November 2025).

withdrawal. Living with Parkinsons disease can lead to stigma, embarrassment, and decline in these can isolate PD patients. Loneliness and inactivity can also worsen symptoms.

Studies show the benefits of music therapy on non-motor quality of life measures were more variable and less definitive than motor outcomes. However, many research studies showed some gains and improvement. The systematic review of 58 papers from 2015 to 2020 by Sotomayor showed many benefits of music therapy to non-motor symptoms. This includes singing, especially group singing, was shown to have some benefit to slow the deterioration of vocal skills. By strengthening the mouth apparatus there was improvement in swallowing and breathing. Improved vocal ability helped communication and social skills. Other studies within the review found that those with PD showed improved motivation, self-confidence, improved motor and gait control, mood regulation, improved focus, emotional expression and improvement of depression after music therapy. Music therapy can play a beneficial role in addressing some non-motor symptoms.

Many PD patients experience insomnia, REM sleep disturbances, and daytime fatigue. Poor sleep worsens mood, cognitive function, and motor control. Impaired sleep can be a significant cause of difficulties for PD patients. Studies show that between 50 to 80% could be affected by poor sleep as a result of the disease³⁶. There has been a suggestion that music therapy could improve sleep in PD patients; however, no studies have yet been conducted to investigate this. Therefore, no solid conclusion can be drawn.

Most studies have overwhelmingly backed the benefit of music therapy on motor function. However, whilst preliminary research on music therapy is promising, improvement in non-motor outcomes will require more studies to provide concrete scientific backing.

Neural plasticity

Neural plasticity is the brain's ability to reorganise structure, functions, and connections³⁷. This functional process allows the brain to form new neural connections. The loss of dopamine disrupts plasticity in motor areas (basal ganglia, motor cortex). Specifically, in PD the brain can re-wire itself to form new pathways to compensate for the

³⁶ Bollu, P.C. and Sahota, P. (2017) Sleep and parkinson disease, Missouri medicine. Available at: <https://pmc.ncbi.nlm.nih.gov/articles/PMC6140184/> (Accessed: 10 October 2025).

³⁷ Neuroplasticity (2025) Wikipedia. Available at: <https://en.wikipedia.org/wiki/Neuroplasticity> (Accessed: 10 November 2025).

loss of dopaminergic neurons. When stimulated by music, the brain may bypass the basal ganglia and use alternative pathways such as those involving the cerebellum and prefrontal cortex, offering a non-pharmacological route to functional improvement ³⁸

Differences in music therapy treatment modalities

The plethora of studies show there are a wide range of modalities and approaches to music therapy that vary in structure and patient involvement. The studies have investigated both active (e.g. singing, instrument playing, or dancing with music) and passive (e.g. listening). Each type can target different motor and non-motor outcomes. The choice and effectiveness of the modality may depend on the goal of the therapy, the individuals' specific challenges with the disease, and the level of participation and engagement required.

Active music therapy requires the patient to create or respond to sound through movement or voice. This includes approaches like singing therapy, percussion exercises, and dance-based therapies. These methods have shown particular benefits in improving vocal strength, respiratory control, and motor coordination. For example, Butala et al ³⁹ demonstrated that group singing improved vocal intensity, breath control, and communication confidence in individuals with PD, suggesting that the active use of voice can counteract hypophonia and social withdrawal. Similarly, Hackney and Earhart found that dance-based therapy (specifically Argentine tango) improved balance, dynamic stability, and movement initiation, outperforming traditional physiotherapy in some trials. These findings suggest that musical experiences with body movements stimulate multisensory integration, reinforcing neural pathways associated with timing, coordination, and posture.

In contrast, passive music therapy, typically involving music listening, requires less cognitive and physical effort but can still yield notable improvements in mood regulation, stress reduction, and sleep quality. Passive modalities are particularly suitable for individuals in advanced stages of PD or those with significant fatigue or cognitive decline.

³⁸ BM, B.I.W.M. (no date) Preliminary neurophysiological evidence of altered cortical activity and connectivity with neurologic music therapy in parkinson's disease, *Frontiers in neuroscience*. Available at: <https://pubmed.ncbi.nlm.nih.gov/30837830/> (Accessed: 10 October 2025).

³⁹ Butala A;Li K;Swaminathan A;Dunlop S;Salnikova Y;Ficek B;Portnoff B;Harper M;Vernon B;Turk B;Mari Z;Pantelyat A; (no date) Parkinsonics: A randomized, blinded, cross-over trial of group singing for motor and nonmotor symptoms in idiopathic parkinson disease, *Parkinson's disease*. Available at: <https://pubmed.ncbi.nlm.nih.gov/36247106/> (Accessed: 10 November 2025).

However, several studies note smaller or less sustained effects on motor symptoms compared to active forms (Buard et al., 2019⁴⁰; Zhou et al., 2023⁴¹).

Some studies have tested combined interventions, where participants engage in active listening alongside physical engagement (e.g. clapping, rhythmic marching). These may offer a balance between accessibility and therapeutic benefit. However, the lack of standardisation across studies, in terms of session length, music selection, or therapist qualifications, makes direct comparison challenging (Li et al., 2023).

The personalisation of music therapy is emerging as a key factor in its effectiveness. When interventions work with a patient's musical preferences, cultural background, and cognitive level, outcomes tend to be more positive. This supports the argument that one-size-fits-all models may be insufficient, and future trials should investigate individualised protocols more rigorously.

Strengths and Limitations of the Evidence Base

The research supporting music therapy for Parkinson's Disease (PD) presents a promising, but inconsistent, evidence base. Many studies show statistically significant improvements in both motor and non-motor outcomes, reliability of these findings vary due to differences in sample sizes, intervention protocols, and measures of outcomes.

There is a growing number of randomised controlled trials (RCTs) and meta-analyses, which provide will higher levels of evidence compared to earlier observational work. For example, Li et al.⁴² (2023) analysed 14 RCTs and found consistent improvements in gait velocity and depression scores ($p < 0.01$), adding further validity to the argument that rhythmic and music-based interventions have measurable therapeutic effects.

However, significant limitations are evident. Many studies include small sample sizes (often less than 25 participants) this reduces statistical reliability and increases the likelihood of variability. This is more evident when investigating non-motor symptoms such

⁴⁰ Braunlich K;Seger CA;Jentink KG;Buard I;Kluger BM;Thaut MH; (no date) Rhythmic auditory cues shape neural network recruitment in parkinson's disease during repetitive motor behavior, The European journal of neuroscience. Available at: <https://pubmed.ncbi.nlm.nih.gov/30375083/> (Accessed: 08 October 2025).

⁴¹ Zhou ZD;Yi LX;Wang DQ;Lim TM;Tan EK; (no date) Role of dopamine in the pathophysiology of parkinson's disease, Translational neurodegeneration. Available at: <https://pubmed.ncbi.nlm.nih.gov/37718439/> (Accessed: 10 October 2025).

⁴² B;, L.H. (no date) Effects of music-based interventions on motor and non-motor symptoms in patients with parkinson's disease: A systematic review and meta-analysis, International journal of environmental research and public health. Available at: <https://pubmed.ncbi.nlm.nih.gov/36673802/> (Accessed: 10 October 2025).

as anxiety or cognitive change, where effect sizes tend to be smaller and more variable. Several RCTs also lack long-term follow-up, raising questions about the sustainability of observed improvements.

Another limitation involves the heterogeneity of intervention design. Studies vary widely in musical genre, tempo, session length, therapist training, and whether interventions are individual, or group based. This lack of standardisation makes cross-study comparison challenging and may partially explain conflicting findings. For example, Sotomayor⁴³ investigated 58 studies each of which had a different treatment protocol.

The use of different outcome measures can limit comparison. Motor outcomes may be assessed using a variety of scoring methods such as UPDRS-III, gait analysis, or observational scoring, while non-motor outcomes often use PDQ-39, PDQ-8, or unvalidated emotional scales. A range of measurement tools introduces variability in meta-analysis.

Another limitation is participant representativeness. Many studies exclude individuals with significant cognitive impairment, advanced PD, or comorbidities, yet these patients are likely to benefit the most from non-pharmacological interventions. Thus, the evidence base may be biased toward younger participants with less advanced PD.

While the literature strongly suggests that music therapy provides benefits, particularly for gait, motor skills and emotional wellbeing, the methodological limitations suggest a clear need for larger, standardised, long-term trials to help the establishment of reliable clinical guidelines and protocols.

Cultural and Demographic Considerations

Although most studies support the effectiveness of music therapy in Parkinson's Disease, few fully explore the effect of cultural, social, or demographic background factors on outcomes. This represents a gap in the literature, given the personal and emotional nature of music.

Most published studies are based in Europe or North America, and often use Western classical or popular music for therapy sessions. While these may be familiar to participants in those regions, they may not have the same cultural relevance for people from other ethnic backgrounds. This could significantly affect emotional engagement, which

⁴³ Machado Sotomayor, M.J. et al. (2021) Music therapy and parkinson's disease: A systematic review from 2015-2020, International journal of environmental research and public health. Available at: <https://pmc.ncbi.nlm.nih.gov/articles/PMC8582661/> (Accessed: 08 December 2025).

in turn impacts motivation, compliance, and therapeutic outcomes (Särkämö et al., 2014⁴⁴).

Few studies have tailored music therapy to participants' personal musical preferences. Evidence suggests that music that is preferred or relevant to the participant enhances dopaminergic activation and emotional response (Salimpoor et al., 2011⁴⁵), and could increase the effectiveness of therapy. Personalisation is particularly important in patients with cognitive decline, where familiarity with music may help memory and reduce anxiety. However, in research settings, there is little evidence of this type of music selection.

Age, gender, and education level may also influence how participants respond to music therapy. Older individuals may find physical modalities like dance more demanding, while younger patients may prefer modern genres or technology-enhanced delivery methods. Digital music is an area that may provide an increased variety and personalised solutions. Many analyses' have limited demographic data which can impact our understanding of how these factors influence the success of therapy.

Most participants in studies have mild-to-moderate impact of PD because they exclude those with advanced disease, cognitive impairment, or comorbidities. This leads to selection bias and raises concerns about the inclusivity of current interventions. For music therapy to be adopted into mainstream practice, future research must reflect the full range of Parkinson's Disease patients and ensure interventions are culturally sensitive, age-appropriate, and suitable across diverse populations.

Combining Music Therapy with Traditional Pharmacological Treatments

Medication remains central to the management of Parkinson's disease. Levodopa is usually offered when motor symptoms affect quality of life, while dopamine agonists and monoamine oxidase-B inhibitors may be considered in some people with earlier or less disruptive symptoms. Other medicines can be added when symptoms fluctuate or the effect of levodopa wears off. Treatment is tailored to the individual because the benefits and adverse effects vary between patients.

⁴⁴ Särkämö T;Tervaniemi M;Laitinen S;Numminen A;Kurki M;Johnson JK;Rantanen P; (no date) Cognitive, emotional, and social benefits of regular musical activities in early dementia: Randomized Controlled Study, *The Gerontologist*. Available at: <https://pubmed.ncbi.nlm.nih.gov/24009169/> (Accessed: 15 October 2025).

⁴⁵ Salimpoor VN;Benovoy M;Larcher K;Dagher A;Zatorre RJ; (no date) Anatomically distinct dopamine release during anticipation and experience of peak emotion to music, *Nature neuroscience*. Available at: <https://pubmed.ncbi.nlm.nih.gov/21217764/> (Accessed: 18 October 2025).

Music therapy should be considered a complementary intervention rather than an alternative to medication. Rhythmic auditory stimulation, singing, dance and other music-based approaches may support gait, speech, mood and quality of life, including symptoms that are not always fully controlled by medication. However, current evidence does not show that music therapy alters the underlying progression of Parkinson's disease or reduces the need for medication.

Few music-therapy studies have systematically monitored medication changes, dosage or interactions with “on” and “off” periods. It is therefore not possible to conclude that music therapy enhances the pharmacological effect of levodopa or allows medication doses to be reduced. Future trials should record medication type, dosage, timing and changes during follow-up so that any relationship between medication and music-based interventions can be assessed more reliably.

Integration with Other Therapies

While music therapy has shown promising effects in managing symptoms of Parkinson's Disease, there is growing recognition of its use when combined with other therapies. This interdisciplinary approach mirrors the complex nature of PD, which affects motor, cognitive, and emotional systems simultaneously. Music therapy could complement and enhance the effects of physiotherapy, occupational therapy, and speech-language therapy.

One of the most cited examples is the combination of Rhythmic Auditory Stimulation (RAS) with gait training or physiotherapy. A systematic review and meta-analysis by Ye et al. (2022⁴⁶) found that adding rhythmic auditory stimulation to physiotherapy produced greater improvements in stride length and gait speed than conventional physiotherapy alone. It also reduced freezing of gait and improved motor function and quality-of-life scores. However, the interventions and control conditions varied between studies, most samples were small, and participants predominantly had mild or moderate Parkinson's disease.

Music therapy also shows potential to complement speech therapy, particularly in addressing symptoms such as hypophonia, dysarthria, and loss of facial expressivity in PD. Activities such as singing and vocal improvisation not only strengthen the vocal cords and breath control but also engage limbic and cortical areas linked to emotional expression and speech production (Tamplin et al., 2019⁴⁷). When music therapy works

⁴⁶ Ye, X., Li, L., He, R., Jia, Y. and Poon, W. (2022) 'Rhythmic auditory stimulation promotes gait recovery in Parkinson's patients: a systematic review and meta-analysis', *Frontiers in Neurology*, 13, 940419. doi: 10.3389/fneur.2022.940419

⁴⁷ Tamplin J;Morris ME;Marigliani C;Baker FA;Vogel AP; (no date) ParkinSong: A controlled trial of singing-based therapy for parkinson's disease, *Neurorehabilitation and neural repair*. Available at: <https://pub-med.ncbi.nlm.nih.gov/31081485/> (Accessed: 11 October 2025).

with traditional speech-language goals, it can help both functional communication and self-expression, especially in patients with social withdrawal.

In occupational therapy (OT), music therapy can help to improve activities of daily living (ADLs) by training fine motor coordination with activities such as playing an instrument, clapping, or rhythmic tasks that simulate real-life activities. According to Buard et al. (2019), these rhythmic and patterned movements help enhance sensorimotor integration, and when combined with OT, could lead to more effective skills in daily routines.

Zhou et al.⁴⁸ (2021) suggest that music-based movement therapy, when combined with tasks requiring executive function (e.g. verbal fluency, working memory), there may be improvements in mental flexibility and motor timing. These findings support the idea that the multi-sensory nature of music therapy may offer neuroprotective or compensatory effects, especially in cognitive and motor domains that may overlap.

Despite these benefits, few studies directly compare these multiple interventions against single ones using standardised testing. This represents a research gap. The available research suggests that music therapy is most effective when used as part of a holistic care plan, tailored to an individual needs and used alongside established treatments.

The integration of music therapy with other therapies has a strong potential for improving functional outcomes and quality of life in PD. Future clinical models should look to interdisciplinary collaboration, working together to share goals, and the customising therapies, ensuring that music is a coordinated part of comprehensive rehabilitation.

Future Directions and Unanswered Questions

Despite a growing body of evidence supporting the use of music therapy in Parkinson's Disease, there are several unanswered questions and gaps in the research that may limit the widespread use of music-based interventions. Future research should look at these areas through better design, diverse participant inclusion, and long-term tracking of results.

Further investigation into the benefit of using music therapy alongside traditional pharmacological treatment options could help to improve the quality of life of PD patients, especially in non-motor symptoms such as wellbeing.

⁴⁸ Zhou Z;Zhou R;Wei W;Luan R;Li K; (no date) Effects of music-based movement therapy on motor function, balance, gait, mental health, and quality of life for patients with parkinson's disease: A systematic review and meta-analysis, Clinical rehabilitation. Available at: <https://pub-med.ncbi.nlm.nih.gov/33517767/> (Accessed: 10 October 2025).

Music therapy may improve sleep in PD patients, further investigation into the effect on sleeping patterns could help to provide relief from those suffering with sleeping difficulties.

One of the most significant limitations in the current literature and studies is the short duration and limited follow-up. Many interventions last only 6 to 12 weeks, with few studies assessing whether improvements in motor or non-motor symptoms are sustained over time. Parkinson's Disease is a progressive, lifelong condition, and for music therapy to be integrated into care pathways, its long-term effectiveness must be shown. Future studies should prioritise longitudinal reviews, with follow-up intervals at 6 months, 1 year, or longer to measure sustainability or decline of benefit.

Another key area for future research is the standardisation of music therapy protocols. Interventions currently vary widely in terms of frequency, duration, music genre, level of patient engagement (active vs passive). While this flexibility allows for personalised care, it also creates challenges in comparing outcomes and generating effective recommendations. Developing a set of evidence-based best practices, personalised by need (e.g. gait vs speech vs mood), would allow for more consistent use in clinical settings.

There is a need for studies that include larger and more diverse populations. Many trials use relatively small samples and often exclude individuals with advanced PD, cognitive impairment, or comorbid conditions. As a result, making general decisions on the findings is limited. Future research should actively include underrepresented groups, older adults, and individuals from non-Western cultural backgrounds, particularly given the personal and emotional nature of music. Understanding how cultural musical preferences, access to therapy, and technology influence outcomes is crucial for effective implementation.

Conclusion

Music therapy is increasingly being recognised as a non-pharmacological intervention for helping to manage both motor and non-motor symptoms of Parkinson's disease (PD). Based on the available evidence, music therapy can be effective in improving specific symptoms such as gait irregularities, speech difficulties, emotional regulation. It can positively help many domains of quality of life, including physical functioning, mental wellbeing, and communication skills.

Motor benefits are most effectively improved with rhythmic-based interventions like Rhythmic Auditory Stimulation (RAS) and dance therapy, which use external cues to improve gait timing, coordination, and posture. These effects are supported by studies using objective measures such as stride length and gait velocity. Vocal and singing-based therapies have similarly shown measurable improvements in voice volume and articulation, helping address decline in speech.

In non-motor symptoms, the evidence base is more variable but still promising. Music therapy has been associated with reduced depression and anxiety, improved mood, and increased social engagement. While the amount of improvement often depends on individual engagement, musical preference, and stage of disease, studies consistently report high acceptance by patients.

However, the overall evidence of effectiveness of music therapy is currently limited by the variability in study design, sample size, and standardisation. Long-term benefits need to be explored more fully. There is insufficient data to confirm whether music therapy can effectively slow the progression of the disease or reduce the need for medication. The use of music therapy in routine clinical care remains inconsistent across regions, often depending on access to trained professionals and funding for this type of care.

Despite these limitations, most studies, including meta-analysis, systematic reviews and randomised controlled trials, suggest that when properly delivered, music therapy offers a safe and engaging intervention that can significantly enhance quality of life in people with Parkinson's disease. Further research with longer follow-up periods and standardised and personalised protocols to deliver therapy will be helpful to fully understanding the therapeutic potential of music therapy.

While not a cure or standalone treatment, music therapy is an effective approach that deserves a more prominent role in the patient-centred care of Parkinson's disease.

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